

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

20ECE43 – DIGITAL IC DESIGN USING VERILOG HDL

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) Differentiate between logical partitioning and physical partitioning by taking an example of a transport monitoring system that checks whether freight trucks drive from one part of the country to another in too short an interval. Draw suitable diagrams.
- b) List and explain different data types in Verilog with an example for each
- c) Draw a block diagram for a negative-edge triggered D flip-flop with clock enable, negative-logic asynchronous preset and clear, and both active-high and active-low outputs. Write the truth table and develop a Verilog behavioral code for the flip flop. It is illegal for both preset and clear to be active together, hence display a statement for this condition.
2. a) Explain how area optimization is achieved at functional design and physical design stages of design flow.
- b) Explain the purpose of MISRs and LFSRs in Built-in Self Test
- c) Write a Verilog code for the Moore FSM having the state diagram given in Fig.Q2(c) in behavioral description.

6 L*2 1 1

6 L2 2 1

8 L3 2 1

6 L2 1 1

6 L2 1 1

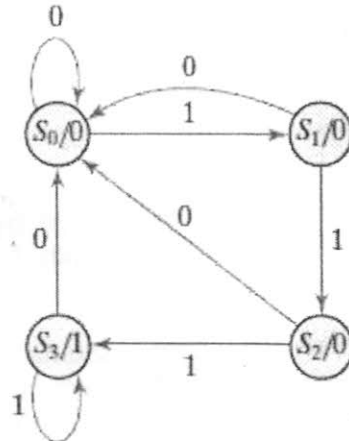


Fig. Q 2(c)

8 L3 2 1

6 L2 1 1

3. a) Explain the steps involved in the physical design of ASIC
- b) Define the following terms
 - i) Code coverage
 - ii) Functional coverage
 - iii) Fault models
 - iv) Fault simulators
 - v) Fault coverages

6 L2 1 1

- c) Write a Verilog behavioral model of the Mealy FSM described by the state diagram in Fig.Q 3(c)

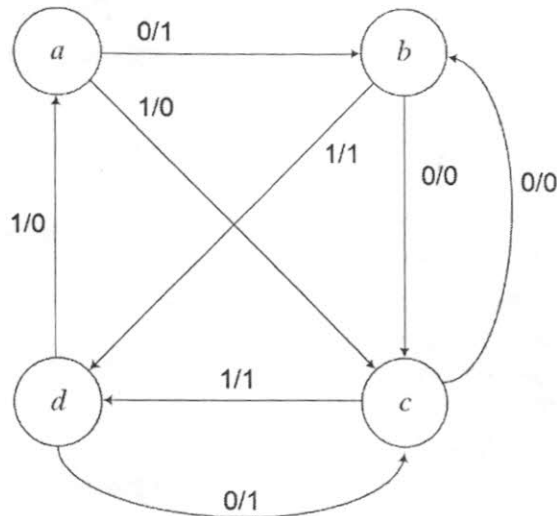


Fig.Q3(c)

8 L3 2 1

Unit – II

4. a) Write a Verilog behavioral code for the four-bit priority encoder described in the following table with the input D_0 having the highest priority and input D_3 the lowest priority along with a valid bit 'V' which is 0 if all the inputs are 0. Also develop a Verilog Testbench for it with four different test cases.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X	Y	V
0	0	0	0	X	X	0
1	X	X	X	0	0	1
0	1	X	X	0	1	1
0	0	1	X	1	0	1
0	0	0	1	1	1	1

10 L3 3 1

- b) Write a Verilog behavioral code for a JK flip-flop with active high reset signal. Also develop a Verilog Testbench for it with four different test cases

10 L3 4 1

5. a) Devise a one-hot code for the state of the traffic light. Assume the code is represented using a 3-element vector with element 1 corresponding to red, 2 to yellow and 3 to green. Develop a Verilog behavioral model for a light controller that has an encoded input, an encoded output, and a single-bit input that enables the lights. When the enable input is 1, the encoded output is the same as the encoded input. When the enable input is 0, all bits of the output are 0. Develop a Testbench model with any four different test cases. Verify the conditions that, when the enable input is 1, the output is the same as the light input, and when the enable input is 0, all light outputs are inactive otherwise display an error message.
- b) Write a behavioral Verilog code for four-bit Ring counter using 'case' statement and develop a Verilog Testbench for it.

10 L3 3 1

10 L3 4 1

6. a) Draw the block diagram for a multiplexer that selects among four sources of data, each of which is encoded with three bits. Develop a Verilog behavioral model and Testbench for it with three different test cases.
- b) Write a Verilog behavioral description for the counter which counts sequence 0,3,5,6,4,7, 0..repeats and develop a Verilog Testbench for it.

10 L3 3 1

10 L3 4 1

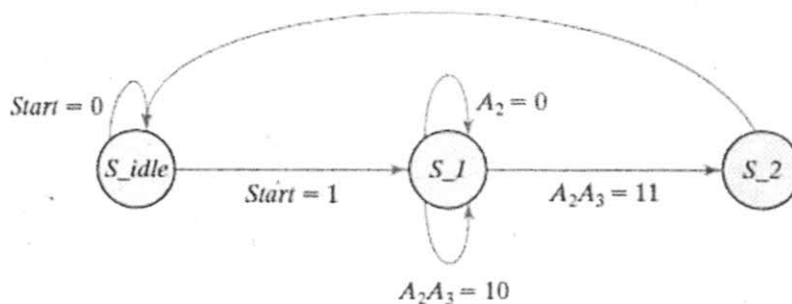
Unit – III

7. a) Explain how the operations in the digital system are categorized with the help of an example for each
- b) Explain the difference between 'for' loop and 'while' loop with the help of Verilog code for each.
- c) A datapath unit consist of two JK flip-flops E and F, and one four-bit binary counter A [3: 0]. The individual flip-flops in A are denoted by A3, A2, A1, and A0, with A3 holding the most significant bit of the count. A signal, 'Start' initiates the system's operation by clearing the counter A and flip-flop F. At each subsequent clock pulse, the counter is incremented by 1 until the operations stop. Counter bits A2 and A3 determine the sequence of operations:
 If A2 = 0, E is cleared to 0 and the count continues.
 If A2 = 1, E is set to 1; then, if A3 = 0, the count continues, but if A3 = 1, F is set to 1 on the next clock pulse and the system stops counting. Then, if 'Start' = 0, the system remains in the initial state, but if 'Start' = 1, the operation cycle repeats. Draw ASMD charts for the following
- controller state transitions, synchronous reset
 - controller state transitions, asynchronous reset.
8. a) Draw an ASMD chart for state diagram and its respective register transfer notation

6 L2 5 1

6 L2 5 1

8 L2 5 1



$S_idle \rightarrow S_1, clr_A_F: A \leftarrow 0, F \leftarrow 0$
 $S_1 \rightarrow S_1, incr_A: A \leftarrow A+1$
 If (A2=1) then set_E: $E \leftarrow 1$
 If (A2=0) then clr_E: $E \leftarrow 0$
 $S_2 \rightarrow S_idle, set_F: F \leftarrow 1$

6 L2 5 1

- b) Explain the operations specified by the following register transfer notation
- $R3 \leftarrow R3 + 1$
 - $R1 \leftarrow R1 + R2, R5 \leftarrow 0$
 - If (T1 = 1) then ($R2 \leftarrow R1$)
 - If (S1 = 1) then ($R0 \leftarrow R1$) else if (S2 = 1) then ($R0 \leftarrow R2$)
- c) Explain the Logic synthesis process with the help of a simplified flowchart for HDL-based modeling, verification, and synthesis.

6 L2 5 1

8 L2 5 1